

Team Number 10

Project Title: Intelligent and Collaborative Autonomous Robots for Real-Time Applications

Report Date: 10/7/2024

Part one: Progress status as a team

All team members have detailed tasks listed on Teams Planner: YES

Please answer the following questions:

1. Have you met as a team this past week? If yes, give date/time, and the members attended the meeting. If no, explain.
 - a. Members did not set a time to all meet this week due to hurricane Milton preparation
2. Have you met with the sponsor as a team? If yes, give date/time, and the member attended the meeting. If no, explain.
 - a. David in the lab on 10/8, Ian on call over teams at 5:30
 - b. Ian in lab on 10/11
3. Describe verbally the tasks completed the past week and the challenges faced.
 - a. A batch file was created in order to localize all the commands needed to run our model on the Unitree Nano. This cuts down on the amount of time entering the same commands every time we perform any tests
 - b. The challenge of the model still being extremely slow has not been solved as yet.
4. Describe the tasks to be completed the coming week.
 - a. Tasks to be completed are to research ways to retain motor function on robot whilst connected to the Internet and run the YOLO model from a personal device.

Copy the following table from the last week's report (9/30)

	# Tasks completed past week	# Tasks not completed	# Tasks for next week
David Uzan	2	0	1
Ian Simmons	3	0	2
John Sinfuego			

Fill in the following table for this week: (10/7)

	# Tasks completed this week	# Tasks not completed	# Tasks for next week
David Uzan	2	0	1
Ian Simmons			
John Sinfuego			

Team Number 10

Project Title: Intelligent and Collaborative Autonomous Robots for Real-Time Applications

Part 2: Individual Progress Report (each member complete one form)

Name of the Member: David Uzan

Report Date: 10/7/24

List the following:

1. All tasks completed the week before the past week (completion date in parenthesis). This is copied from the last report, and if this report is the first one, skip.
2. All tasks completed the past week (completion date in parenthesis)
3. All tasks you are currently working on or planned for this coming week (completion date in parenthesis)

For **each** task you have completed **during the before the past week**:

1. Task number/name

Personal Week 5 (9-30-24) - Add object detection python file to Uni-Go1

Personal Week 6 (9-30-24) - Run and display live tracking from robot

For **each** task you have completed **during the past week**:

1. Task number/name

Personal Week 7 (10-7-24) - Create Batch file to kill camera processes and run python file on dog. Run on dog while in motion.

2. Short description – these descriptions form the basis of the final project report
 - a. Implementation, testing, results

A batch file is very easy to create. I found the commands to kill the camera processes on the robot in the history of the nano responsible for the front camera. I compiled them and called to run the python file containing the object detection model which is intended to be run when the robot is on a fresh restart. There is no risk of running the batch file again, it would simply make more sense to run the python file alone because the camera processes would already be killed.

The robot has capabilities to connect to the Internet with ethernet, but doing so would release its joints and bind it to the floor unable to move. I made a batch file to connect it to the internet on the raspberry pi, but to make the robot move I kept it offline.

b. Brief and to point

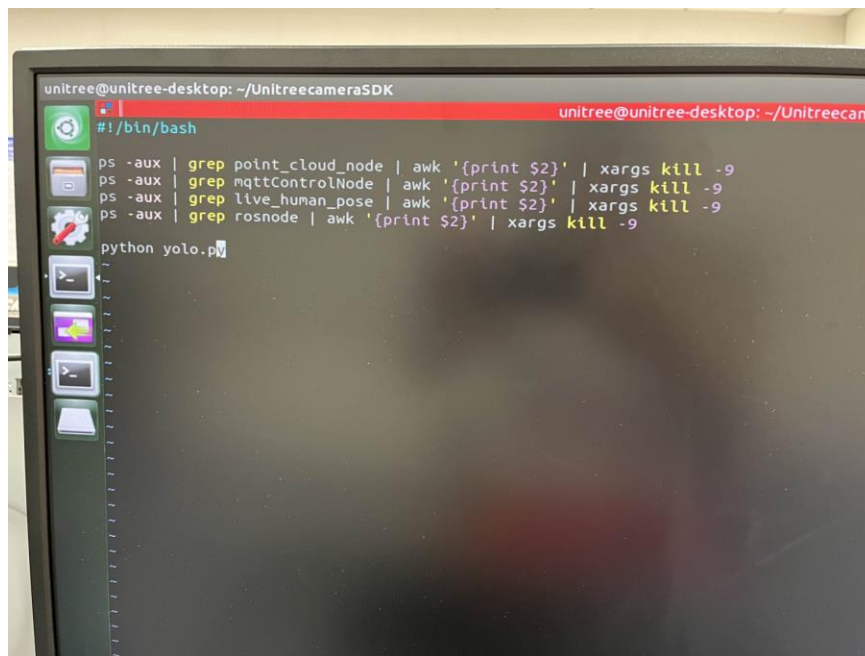
The batch file was simple to create and works as intended. The robot can give live feed while moving but it must not be connected to the Internet.

3. Outcome of the task activity

a. How did the task advance your project?

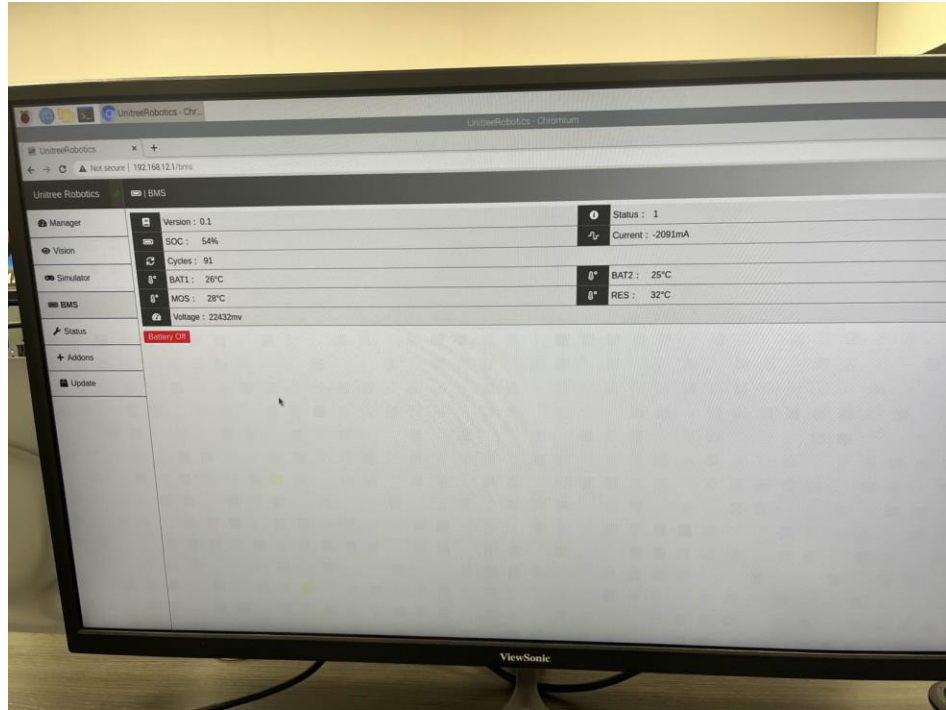
Instead of having to run long and in-depth commands we can save time running the batch file to connect to the robot's cameras. The live feed from the robot also shows that it is capable of doing so without being physically connected to the monitor and peripherals, all we need to do is find how to not lose motor function with Internet connection.

b. Include screen shots, link to videos, plots, data tables, designs, figures, etc. as evidence of your work on the task



```
unitree@unitree-desktop: ~/UnitreecameraSDK
#!/bin/bash
ps -aux | grep point_cloud_node | awk '{print $2}' | xargs kill -9
ps -aux | grep mqttControlNode | awk '{print $2}' | xargs kill -9
ps -aux | grep live_human_pose | awk '{print $2}' | xargs kill -9
ps -aux | grep rosnod | awk '{print $2}' | xargs kill -9
python yolo.py
```

Created the bash file for the camera sdk that kills camera processes so the model can run.



Check the status and data points of the robot. Delay of about 15 seconds.

Below is the link for a video of the robot moving while running the object detection model.

<https://youtube.com/shorts/32m1Ja79M88?feature=share>

For **each** task you are currently working on or plan to work on in the next period:

1. Task number/name

Personal Week 8 (10-13-24) - Research means to keep motor function with Internet and run model from the robot on a personal computer.

2. Short description

a. Plans, resources needed, implementation, results

To allow for practicality of more applicable use cases, we must test the ability to run the models off personal computers that do not have the same level of access to the robots inside system ware.

b. Main challenges, reasons for any delay in completion

There may be a disconnect between the personal computer's terminal and the onboard terminals.

3. Outcome of the task activity

- a. Include screen shots, link to videos, plots, data tables, designs, figures, etc. as evidence of your work on the task

N/A

Part 2: Individual Progress Report (each member complete one form)

Name of the Member: John Angelo Sinfuego

Report Date: 10/7/24

List the following:

1. All tasks completed the week before the past week (completion date in parenthesis). This is copied from the last report, and if this report is the first one, skip.
2. All tasks completed the past week (completion date in parenthesis)
3. All tasks you are currently working on or planned for this coming week (completion date in parenthesis)

For **each** task you have completed **during the past week**:

Part 2: Individual Progress Report (each member complete one form)

Name of the Member: Ian Simmons

Report Date: 10/7/2024

List the following:

1. All tasks completed the week before the past week (completion date in parenthesis).
This is copied from the last report, and if this report is the first one, skip.
2. All tasks completed the past week (completion date in parenthesis)
3. All tasks you are currently working on or planned for this coming week (completion date in parenthesis)

For **each** task you have completed **before the past week**